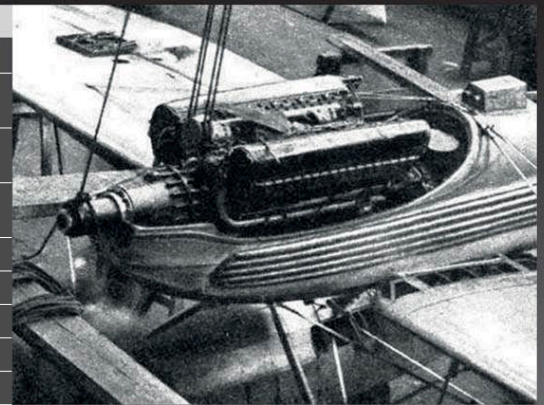


ENGINE SPECIFICATIONS

Dates produced	1929-1931
Configuration	Water-cooled supercharged 60-degree V12
Fuel	Leaded benzole/methanol/acetone mixture
Power output	2,530 hp @ 3,200 rpm, ultimately 2,783 hp
Weight	1,640 lb (744 kg) dry
Displacement	2,239 cu in (36.7 litres)
Bore and stroke	6 in x 6.6 in (167.6 mm x 152.4 mm)
Compression ratio	6.0:1

► See Piston engines pp.302-303

**Spark plugs**

The outer set only is visible—the second set is on the inside of the “V” of the cylinders.

Exhaust ports

The R had four-valve cylinder heads, the two exhaust valves sharing a common exhaust port (covered for display).

Camshaft drive shaft
(enclosed in tube)**Inlet manifold**

Because the engine was run rich—with excess fuel in the air/fuel mixture—the cooling effect of the evaporation of fuel droplets made an intercooler unnecessary.

Magneto

This is one of a pair supplying ignition to dual sets of spark plugs.

Double-sided supercharger

This runs at eight times engine speed.

Downdraft air intake

This is designed to avoid water spray on takeoff.

Shallow dry sump

Oil is circulated to a tank in the S.6's tailfin, via corrugated coolers built into the fuselage sides.

Water pump

Coolant is circulated to radiators that form the surface of the Supermarine S.6's wings or floats.

